

Focused EDA for ID/OOD Scenario Design (M5 Forecasting)

This report characterizes the temporal demand behavior of the M5 dataset across predefined, chronologically structured evaluation windows. The analysis supports the thesis methodology and results sections, establishing the justification for the **OOD temporal scenario** (or **temporally shifted retail evaluation scenario**) without claiming any severe or extreme distribution shift.

Predefined Evaluation Windows

By chronological design, the dataset is partitioned into four distinct periods. This structure preserves the temporal ordering of data, avoids leakage, and establishes evaluation windows matching the standard M5 forecasting horizon of 28 days:

Window	Day Range	Start Date	End Date	Duration (Days)	Description
Train	d_1 – d_1857	2011-01-29	2016-02-28	1857	Long-term historical baseline
Validation	d_1858 – d_1885	2016-02-29	2016-03-27	28	Near-future tuning phase
ID Test	d_1886 – d_1913	2016-03-28	2016-04-24	28	Ordinary near-future test window
OOD Test	d_1914 – d_1941	2016-04-25	2016-05-22	28	Temporally shifted retail scenario

Predefined Window Characterization

The window statistics are calculated across four primary signals: event exposure/calendar intensity, volatility, zero-sales intermittency, and pricing activity.

Summary Comparison Table

Metric	Train (d_1 – d_1857)	Validation (d_1858 – d_1885)	ID Test (d_1886 – d_1913)	OOD Test (d_1914 – d_1941)
Agg. Daily Mean Sales	34,106.47	42,002.39	42,272.36	43,991.57
Agg. Daily Median Sales	33,518.00	39,884.50	40,050.50	42,724.00
Agg. Daily Std. Dev.	7,232.88	6,569.96	7,220.30	6,759.78
Agg. Daily CV	0.2121	0.1564	0.1708	0.1537
Mean SKU-level CV	2.4433	1.6062	1.6157	1.6302
Zero-Sales Ratio	0.6854	0.5741	0.5628	0.5444
Active SKU Ratio	1.0000	0.9295	0.9480	0.9734
Event Frequency	0.0813	0.1071	0.0000	0.1429
SNAP Frequency	0.3285	0.3571	0.3571	0.3571
Holiday Count	102	1	0	2
Avg. SKU Price CV	0.0321	0.0021	0.0011	0.0016
Price Change Freq.	0.0017	0.0022	0.0008	0.0010

Characterization of the OOD Temporal Scenario

The OOD test window (d_1914 to d_1941) displays distinct, predefined signals that distinguish it from the ordinary ID test window (d_1886 to d_1913), justifying its role as a **temporally shifted retail evaluation**

scenario:

1. Event Exposure & Calendar Intensity:

- The **ID Test window is completely event-free** (Event Frequency = 0.0000 , Holiday Count = 0).
- The **OOD Test window has high calendar intensity** (Event Frequency = 0.1429 , Holiday Count = 2). It spans several notable events including Orthodox Easter, Cinco de Mayo, and Mother's Day, which typically trigger shifts in shopping behaviors and demand spikes.

2. Pricing Activity:

- The price change frequency in the OOD window is **35.1% higher** than in the ID window (0.001037 vs 0.000768), indicating heightened retail promotional activity.
- The average SKU-level price variation (CV) is **35.8% higher** in the OOD window (0.001551 vs 0.001142), reflecting dynamic price shifts during the event-heavy period.

3. Demand Volatility & Sales Activity:

- While the aggregate daily CV in OOD is slightly lower (0.1537 vs 0.1708), the average individual SKU-level volatility (Mean SKU CV) is slightly higher (1.6302 vs 1.6157).
- The OOD window exhibits a higher SKU active ratio (0.9734 vs 0.9480) and a lower zero-sales ratio (0.5444 vs 0.5628), indicating a period of broader purchase engagement across the catalog.

Descriptive Shift Metrics (vs. Training Window)

To further document the shift characteristics, we present standard distribution shift metrics compared to the historical baseline (Training Window):

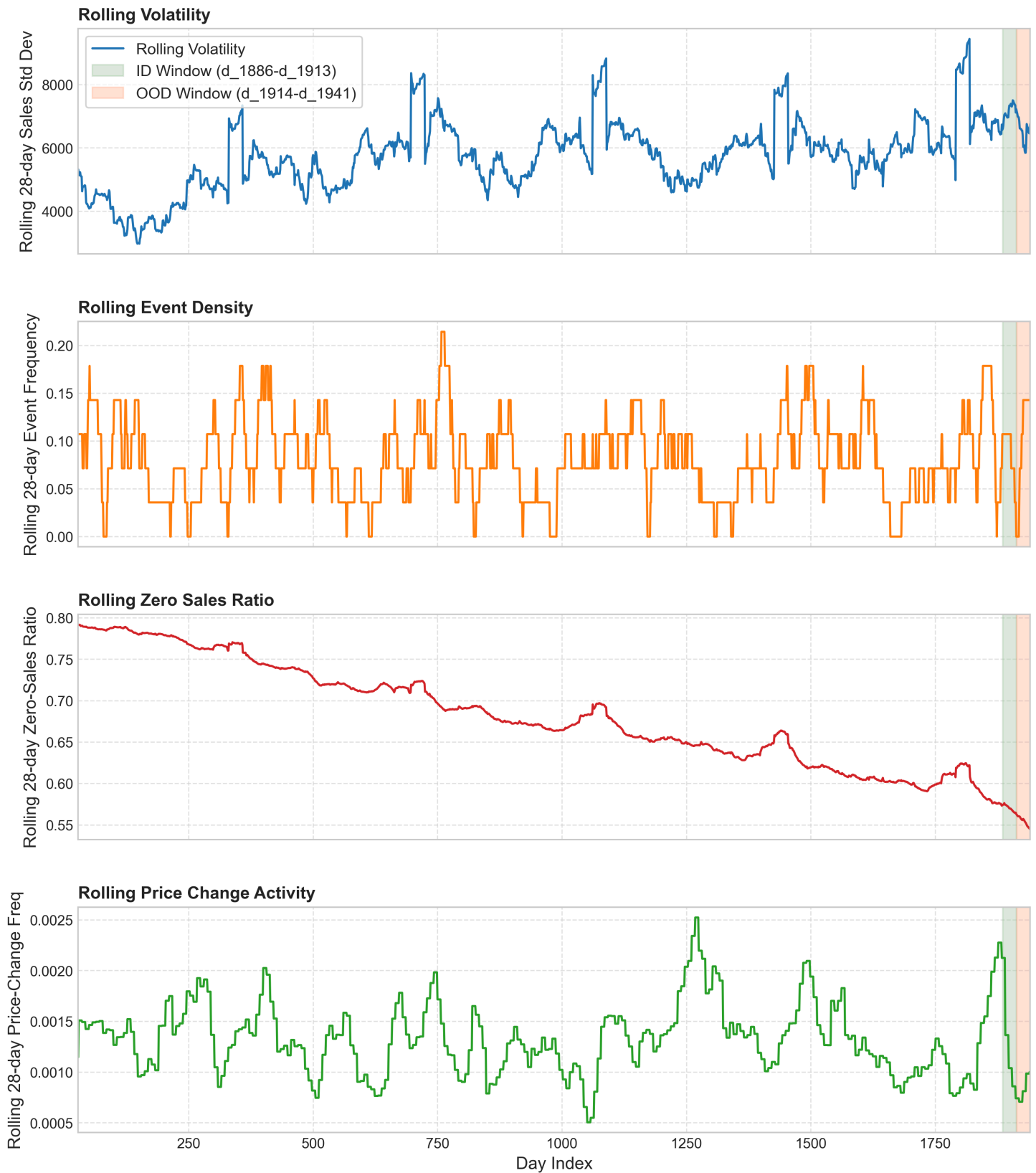
Comparison (vs. Train)	Level	Mean Diff.	Var. Diff.	Rel. Volatility Diff.	KS Statistic	Wasserstein Dist.
Train vs. ID Test	Aggregate (Daily Sum)	+8,165.89	-181,942.19	-0.17%	0.5042	8,165.89
Train vs. OOD Test	Aggregate (Daily Sum)	+9,885.10	-6,619,947.14	-6.54%	0.5839	9,885.10
Train vs. ID Test	SKU-level (Subsample)	+0.2606	-3.62	—	0.2070	0.7848
Train vs. OOD Test	SKU-level (Subsample)	+0.3081	-3.30	—	0.2097	0.7869

Note: SKU-level statistics represent the average across a representative random subsample of 3,000 series. At the aggregate level, the OOD test window exhibits a larger Kolmogorov-Smirnov (KS) statistic (0.5839 vs 0.5042) and a greater Wasserstein distance (9,885.10 vs 8,165.89) relative to the training baseline, matching the descriptive shift behavior.

Visualizations

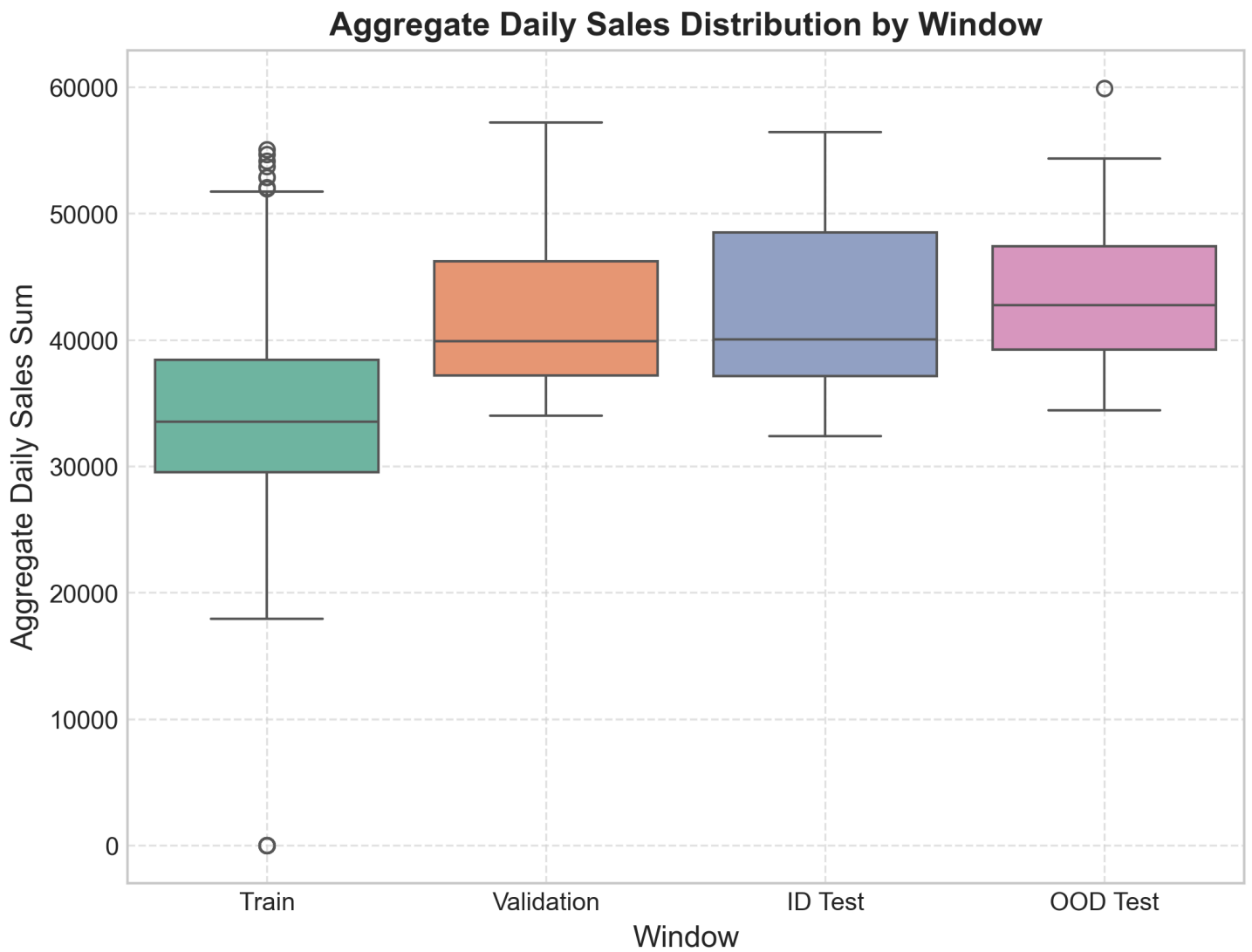
1. Rolling Temporal Metrics

This dashboard displays rolling 28-day temporal statistics over the entire historical range (from day 28 to day 1941). The ID and OOD windows are shaded in green and red respectively, illustrating how the metrics evolve dynamically over time:



2. Aggregate Daily Sales Distributions

This boxplot compares the daily sales sum distribution across all four chronological partitions, illustrating the shift in mean daily demand:



3. Window Characterization Metrics

These bar charts compare key volatility and calendar/price density indicators, highlighting the event exposure and pricing changes characterizing the OOD window:

